

Gating Benefits Depend on the Backbone: An Ablation Study of a Mamba-3 Block under Matched-Noise Cross-Condition Drift

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Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run (internal pre-specification, not an external registry). First, replacing the input-dependent parts of Δ , A , B and C by learned constants—while the input-dependent trapezoidal weights and rotation angles of the Mamba-3 scan, the convolutional stem and the output gate are kept—costs at most about 5 pp under deep noise (paired intervals include zero), and the frozen block still outperforms a parameter-matched time-invariant S4D by 14–20 pp. Under the pre-specified criterion these four projections are therefore not the primary contributor. A second grid then freezes the trapezoidal weight and the rotation angles as well, leaving no tensor entering the scan dependent on the input: the advantage over S4D does not shrink but measures +21.9, +24.0 and +21.2 pp at 0, -2 and -6 dB (5/5 seeds, intervals excluding zero), with fewer parameters, so the block’s deep-noise advantage over a time-invariant SSM requires no input-dependent scan term at all. Second, removing the multiplicative SiLU gate from the frozen block costs 9–24 pp and returns it to roughly the level of S4D. Third, adding the gate to S4D recovers part of the gap, but not because of the gate: a narrow graft recovers 17–20%, a graft

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rebuilt in the donor’s width and projection structure recovers 30% (seed interval 16–52%; the pre-specified verdict is inconclusive between insufficient and partial recovery), and an additive branch with identical parameters recovers at least as much (41%). In the S4D host the multiplicative gate thus adds nothing beyond an equal-capacity additive branch, while its removal from the Mamba-3 block is costly; in this sense the gate’s benefit depends on the host. On the reverse transition (the same bearings with their roles swapped) the fully input-independent scan again exceeds S4D by 22–26 pp and gate removal again costs 18–24 pp, all intervals excluding zero. A separate, shorter per-class grid shows the gate-related differences concentrating on the minority inner-race class. Values are best-epoch macro-F1 on the evaluation condition; the central contrasts for four arms are re-assessed under last-epoch reporting. The evidence is conditional on one block, one pair of operating conditions run in both directions with the same eight bearings, five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet it sits in unresolved tension with recent evidence: the first systematic S4D-versus-Mamba comparison on time-series classification finds the *simpler, time-invariant* model consistently superior [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells (hash-verified), and the block can be

19 dissected.

20 Our approach is deliberately *attributional* rather than *methodological*: we
21 propose no new architecture and claim no state-of-the-art. Instead we take
22 a regime in which a Mamba-style block exhibits a large, stable robustness
23 advantage over a time-invariant SSM, and ask which ingredient earns it.
24 The chain has three steps, each with adjudication thresholds fixed before the
25 corresponding grid was run:

- 26 1. **Test the classical selectivity terms (necessity test).** We con-
27 struct a *frozen* arm: the input-dependent parts of Δ , A , B and C
28 are replaced by learned constants. The Mamba-3 scan also takes an
29 input-dependent trapezoidal weight and input-dependent rotation an-
30 gles; these, the convolutional stem, the multiplicative SiLU gate, the s-
31 tate parameterisation, normalisation and projections are retained (Sec-
32 tion 3.2). If these four terms drive robustness, the frozen arm should
33 collapse toward the time-invariant baseline.
- 34 2. **Localise the necessary component.** On the frozen base, we remove
35 one component at a time (convolutional stem; SiLU gate; heavy-tailed
36 state parameterisation) under a single-variable discipline checked key-
37 by-key on the state dictionary (Section 3.2), and adjudicate each arm
38 against pre-specified Δ -thresholds.
- 39 3. **Test sufficiency by constructive grafting.** The component iden-
40 tified as necessary is added to plain S4D as a gate branch, and a pre-
41 specified recovery ratio decides whether “removal collapses” is matched
42 by “addition restores”.

43 The outcome is a conditional pattern. Freezing the input-dependent Δ ,
44 A , B , C projections costs at most about 5 pp in deep noise (paired intervals
45 include zero) while the block retains a 14–20 pp advantage over S4D (best-
46 epoch and last-epoch reporting, Section 4.7): under the pre-specified criterion
47 these projections are not the primary contributor. Freezing the trapezoidal
48 weight and the rotation angles as well—so that nothing entering the scan
49 depends on the input—does not shrink the advantage either: it measures
50 +21 to +24 pp over S4D with fewer parameters (Section 4.2). Removing the
51 SiLU gate costs 9–24 pp and takes the block to roughly the level of plain
52 S4D or below. Adding a gate branch to S4D recovers only $R = 0.17$ of
53 the gap (0.20 under last-epoch reporting). Rebuilt in the donor’s width
54 and projection structure, the graft recovers $R = 0.30$ (interval 0.16–0.52),

55 but an additive branch with identical parameters recovers at least as much
 56 (0.41, Section 4.5). In the S4D host, then, the multiplicative gate adds
 57 nothing beyond an equal-capacity additive branch, whereas in the Mamba-3
 58 block its removal is costly; we read this as a host dependence of the gate’s
 59 benefit in this design. The first two steps replicate on the reverse transition
 60 (Section 4.6). A per-class analysis on a separate, shorter grid indicates that
 61 the gate-related differences concentrate on the minority class.

62 **Contributions.** (i) A layered freezing experiment on a Mamba-3 block
 63 under distribution drift and matched noise, from the classical selectivity
 64 terms (Δ , A , B , C) through to a scan in which no tensor depends on the
 65 input, in which the block’s deep-noise advantage over S4D survives every
 66 step; each arm is verified input-independent by capturing the tensors the
 67 scan receives for two different inputs. (ii) Component-level evidence that the
 68 output gate is the one ablated component whose removal causes a large, one-
 69 signed loss, and that adding it to a time-invariant host, even in the donor’s
 70 width and with an additive control of identical size, recovers no more than
 71 the added structure does; both of the first two findings replicate on the re-
 72 verse transition. (iii) A reporting-rule sensitivity analysis (best epoch on the
 73 evaluation condition versus final epoch) showing that absolute levels shift by
 74 1.4–9.5 pp while the central contrasts keep their sign. (iv) A documented
 75 pre-specification and review trail—evaluation-label hashes, thresholds writ-
 76 ten before each grid, and agent-assisted review—including the incidents in
 77 which automated adjudication departed from the pre-specified criteria; we
 78 release the trail.

79 2. Background and Related Work

80 2.1. Time-invariant and selective SSMs

81 Diagonal structured SSMs (S4D) [6] evolve a linear state with input-
 82 independent parameters (A , B , C , Δ), acting as a bank of learned linear
 83 filters. Mamba-style blocks [1] make Δ , B , C functions of the input token
 84 (selectivity), wrap the scan in a block containing a short causal convolu-
 85 tional stem, a multiplicative SiLU gate on a parallel branch, and residu-
 86 al/normalisation structure; the concrete block used throughout this paper
 87 follows the Mamba-3 formulation [7]. The narrative crediting selectivity for
 88 robustness draws on theory showing input-dependent dynamics can represent
 89 discontinuity-sensitive projections and counteract memory decay [8], and on

[9]. We treat these as *hypotheses to be tested*, not established attribution-
s: the theoretical results concern representability, which does not by itself
determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic 59-dataset comparison finds S4D-based models superior to Mamba variants on clean time-series classification [4]; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change

(cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.8). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.8 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every reported cell carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication. The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and three follow-up grids add 225 cells (Sections 4.2, 4.7, 4.5 and 4.6).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty. One follow-up grid (Section 4.6) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the same eight bearings with their roles swapped, so it tests the other direction rather than independent bearings; its evaluation set holds 4,624 OR and 624 IR windows, all IR windows coming from Bearing2_1, and its label hash is 93424834.... The 35 Hz/12 kN condition has no inner-race bearing and cannot form a binary pair.

The principal arms are summarised in Table 1.

3.2. What the frozen arm freezes, and what the checks verify

In the Mamba-3 layer used here, one input projection emits the gate and value streams, the scan inputs B and C , the step-size and decay terms (Δ and A , through input-dependent additive terms), a trapezoidal weight and

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: no tensor entering the scan depends on the input	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
CNN/TCN baselines	Conventional baselines for context	—

161 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 162 A , B and C and replaces them with learned constants broadcast over time
 163 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 164 rameter plays the role of a static log-decay, passed through the layer’s own
 165 heavy-tailed clamp). The removed projections are deleted, so no parameter
 166 or gradient can leak. **The trapezoidal weight and the rotation angles**
 167 **remain functions of the current input**, as do the stem and the output
 168 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 169 formulation [7] the input-dependent rotation and trapezoidal weighting also
 170 enter the state update. Section 4.1 is accordingly a statement about the
 171 classical selectivity terms Δ , A , B , C only. Three further arms close that gap
 172 by the same device: *frozen-trap* replaces the trapezoidal weight by a learned
 173 per-head constant, *frozen-rot* replaces the rotation angles by learned con-
 174 stants (so the rotation becomes a learned position-only phase rather than a
 175 function of the current input), and *frozen-all* does both, leaving **no tensor**
 176 **entering the scan dependent on the input**. In each case the correspond-
 177 ing `in.proj` output slice is deleted. Section 4.2 reports them. For these arms
 178 the construction check adds the decisive probe: the tensors handed to the
 179 scan are captured for two different inputs through the same module, and a
 180 tensor the arm claims to have frozen is required to be bitwise identical across
 181 them while a tensor left selective is required to differ (all three arms pass).
 182 No kernel modification is needed, because the scan interface accepts these
 183 tensors regardless of their origin.

184 Single-variable discipline was checked by key-level state-dictionary diffs
 185 between two models built under the same seed. For the frozen arm the
 186 diff verifies that every non-target tensor has the same name and shape as in
 187 BM3; for the component-ablation and graft arms it additionally compares the
 188 values of non-target tensors at initialisation. Every differing key is mapped
 189 to an entry in an architecture map. We do not claim byte-level identity of
 190 non-target tensors in the frozen arm, and trained weights naturally differ
 191 between arms; the checks concern how the arms were constructed.

192 3.3. Reporting rule

193 The macro-F1 reported for each cell is the *maximum over epochs* of macro-
 194 F1 evaluated on the evaluation split, i.e., the held-out operating condition
 195 that every arm is scored on (the split covered by the fingerprint above).
 196 The harness defines no separate validation split, so the epoch is chosen with
 197 the evaluation data. Two consequences follow. First, absolute values are

198 optimistically biased, by an amount that grows with an arm’s epoch-to-epoch
 199 variability. Second, a common rule cancels a shared offset between arms but
 200 not an arm-dependent one. The pre-specified thresholds were fixed against
 201 this rule, so all adjudications in Section 4 are reported under it, unchanged.
 202 Section 4.7 then re-assesses the central contrasts under a rule that does not
 203 use the evaluation split (the final epoch), and the boundaries section states
 204 what that re-assessment does and does not cover.

205 3.4. Confound elimination

206 Before component attribution, three candidate confounds were examined
 207 on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D
 208 pointwise (mean -0.5 pp), so this particular increase in S4D’s parameter
 209 count does not explain the gap; S4D-wide was not re-run under last-epoch
 210 reporting, and the control does not address the change in channel mixing
 211 that gate removal also entails. **Normalisation:** both SSM arms use Layer-
 212 Norm and contain no BatchNorm; the arm that degrades under noise is the
 213 one *without* any batch statistics, which argues against BatchNorm statistic-
 214 s mismatch as the explanation. Other normalisation inside the Mamba-3
 215 layer was not varied. **Auxiliary physics loss:** an exploratory kinematics-
 216 alignment regulariser carried by the original model contributes -0.1 pp point-
 217 wise (curves coincide); all attribution arms are therefore built on the pure
 218 cross-entropy variant.

219 3.5. Pre-specification and adjudication

220 Every adjudicated question in this study had numeric decision thresholds
 221 written to a task card and to disk before the corresponding grid was launched.
 222 This is internal pre-specification, not external registration: the supporting
 223 evidence is local file modification times and run-directory names (released,
 224 [Appendix A](#)), which are not tamper-evident and are not preserved by a
 225 repository checkout, so the trail lets readers inspect the order of events but
 226 cannot prove it. The exception is the last follow-up grid (Sections 4.5–4.6):
 227 its thresholds were committed to the public repository before its first cell was
 228 run, which gives that grid an external time-stamp. Later questions (compo-
 229 nent ablation, graft) were specified after earlier grids had been seen, as is
 230 natural in a staged design. Adjudication was mechanical: an analysis pass
 231 applies the thresholds and emits a decision artifact; an independent reviewer
 232 pass re-computes aggregates, verifies fingerprints row-by-row, and must cite

file paths and modification times for every check. The pre-specified thresholds were binding in practice: two attempted post-hoc criterion substitutions and one stale-artifact false verification were caught and re-adjudicated under the original rules — this is the one place where pre-specification actually bit. [Appendix C](#) details the incidents, the automated executor/reviewer implementation, and the human adjudication gates; all scientific decisions — thresholds, rulings, and interpretations — were made by the authors and are timestamped. Statistical tests (paired, per-condition) are reported throughout but were never adjudication criteria. A single repeat of one 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; [Appendix B](#)); this is an observed repeat difference, not an estimated variability bound, and arm-level contrasts are judged against seed-level variability (2–9 pp).

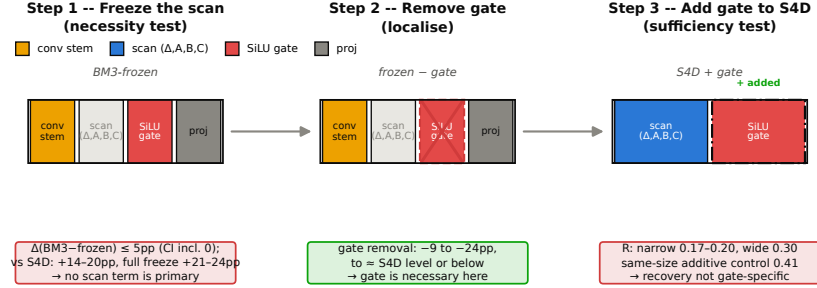
4. Results

Throughout, “deep noise” denotes the pre-specified adjudication band $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-invariant baseline is near chance there: a constant majority-class predictor scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this evaluation set, against 52.3% for S4D). All values are macro-F1 (%), mean over 5 seeds, best-epoch on the evaluation condition (Section 3.3) unless stated otherwise. Figure 1 summarises the three-step ablation chain that structures this section.

4.1. Freezing the input-dependent Δ , A , B , C projections

Table 2: BM3 vs. BM3-frozen vs. S4D, macro-F1 (%) by SNR (frozen-selectivity grid). $\Delta(\text{BM3} - \text{frozen})$ and $\Delta(\text{frozen} - \text{S4D})$ in percentage points.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3} - \text{frozen})$	$\Delta(\text{frozen} - \text{S4D})$
clean	78.9	78.9	81.0	−0.0	−2.1
0 dB	95.0	89.9	76.1	+5.1	+ 13.8
−2 dB	95.4	92.1	75.1	+3.3	+ 17.0
−6 dB	88.5	86.3	68.5	+2.2	+ 17.7
−10 dB*	68.1	64.3	52.3	+3.8	+12.0



Conditional on this block, testbed and budget: the gate's benefit depends on the host.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.3), and graft that ingredient onto S4D to test sufficiency (Section 4.4).

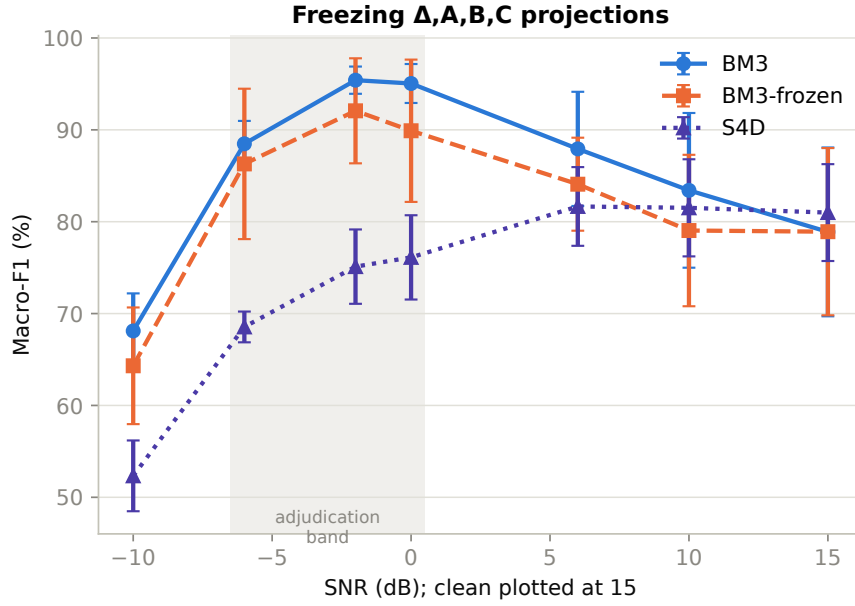


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid. The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs at most 5.1 pp anywhere in the adjudication band on the point estimates, while the frozen block retains a 13.8–17.7 pp advantage over S4D (Table 2, Figure 2). Five seeds bound these quantities loosely: paired 95% intervals for $\Delta(\text{BM3-frozen})$ are $[-4.7, 15.0]$, $[-3.1, 9.8]$ and $[-7.5, 11.8]$ pp at 0, -2 and -6 dB (BM3 ahead in 3/5, 4/5 and 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ $[-0.5, 28.0]$, $[5.5, 28.4]$ and $[5.8, 29.7]$ pp (5/5, 5/5, 5/5 seeds). The pre-specified rule attributed robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in deep noise with the frozen arm collapsing toward S4D. Neither condition holds on the point estimates, so the pre-specified verdict is that these projections are *not the primary mechanism*. This is a failure to meet a confirmation criterion, not a demonstration that freezing is cheap: the intervals above are compatible with differences larger than 10 pp, and no equivalence margin was pre-specified or tested. One limit applies: the frozen arm still contains input-dependent trapezoidal weights and rotation angles, so this result concerns Δ , A , B , C only; Section 4.2 removes that limit. Under last-epoch reporting the re-run gives $\Delta(\text{BM3-frozen}) = +3.7$, $+0.9$ and $+0.2$ pp (best-epoch re-run: $+5.1$, $+3.5$, $+3.0$ pp; intervals still include zero at every level), so the point estimates fall further below the 10 pp threshold rather than rising towards it. On clean data S4D matches or slightly exceeds both Mamba-style arms—consistent with the systematic clean-TSC comparison [4]. In the noisy regime, the block’s advantage over S4D is not removed by freezing these four terms, so it must be carried by other parts of the block or by the remaining input-dependent scan terms.

4.2. Freezing the remaining input-dependent scan terms

Table 3 reports the three layered arms of Section 3.2 on the adjudication band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training code, the same evaluation fingerprint, with the original BM3 arm re-run in the same grid). The question is whether the advantage that survived freezing Δ , A , B , C is carried by the two input-dependent terms the frozen arm retained.

It is not. With every tensor entering the scan input-independent, *frozen-all* is $+21.9$, $+24.0$ and $+21.2$ pp above S4D at 0, -2 and -6 dB under final-epoch reporting, positive in 5/5 seeds at every level and with intervals excluding zero—if anything a larger margin than the partially frozen arm’s $+15.5$, $+19.8$, $+16.4$ pp (Table 9). None of the three freezing steps costs accuracy: the paired differences $\text{BM3-frozen} - \text{frozen-trap}$, $-\text{frozen-rot}$

Table 3: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen−trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen−rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen−all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen−all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen−all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen−all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

and − frozen−all are negative at all three levels (between −1.8 and −6.3 pp under final-epoch reporting), i.e. the more heavily frozen arm is numerically ahead, with every interval including zero. The direction is not an artefact of capacity: *frozen−all* has 103,842 parameters against 112,410 for BM3-frozen and 177,938 for BM3. At the deepest level *frozen−all* also exceeds the full BM3 block by 4.5 pp ([−7.2, −1.9] for BM3 − frozen−all, 0/5 seeds in BM3’s favour); we report this but do not build on it, since five seeds and one speed pair cannot separate a genuine advantage of a time-invariant scan from a variance effect.

Two readings of the whole step are therefore available, and the data do not separate them: input-dependence of the scan contributes nothing measurable here, or it contributes something that the learned constants recover under this training budget. Either way the conclusion for the attribution chain is the same and now rests on a complete intervention rather than a partial one: **the Mamba-style block’s deep-noise advantage over a time-invariant S4D does not require any input-dependent term in its scan**. What remains to carry it is the rest of the block—the convolutional stem, the heavy-tailed A parameterisation, the normalisations and the output gate—which is what Sections 4.3–4.4 examine. The re-run also reproduces the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed Table 2’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build,

314 Section 4.7).

315 4.3. Removing the gate is the only ablation with a large, one-signed effect

Table 4: Single-component removal on the frozen base (component-ablation grid, trained separately from Table 2), $\Delta(\text{frozen} - \text{arm})$ in percentage points.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+ 13.1	+ 21.1	+ 23.5	PRIMARY (all ≥ 8 pp)
conv stem ($k=7$)	−6.6	−2.7	+7.9	outside pre-specified bins (regime trade-off)
heavy-tailed $A \rightarrow$ S4D-style A	−2.4	−2.2	−4.2	outside bins (removal numerically beneficial; intervals include 0)

316 “Necessary” is used here in the operational sense of this ablation: remov-
317 ing the gate from the frozen block, with the other components and the budget
318 unchanged, removes most of the block’s advantage. The gate result is large,
319 one-signed and monotone in noise depth: removal costs 13.1, 21.1 and 23.5 pp
320 at 0, −2 and −6 dB (paired 95% intervals [6.5, 19.6], [3.8, 38.5], [8.5, 38.6] pp;
321 5/5 seeds at every level). The gateless arm also gives up most of the Mamba
322 advantage: it sits **below plain S4D** at −2 dB (71.7 vs. 75.2) and −6 dB
323 (62.5 vs. 68.6), though not at 0 dB (77.6 vs. 76.2), and the gateless–S4D gap
324 itself is not resolved by five seeds (Section 4.7: under last-epoch reporting
325 the gateless arm is below S4D at −6 dB only). We read this as “at best on
326 par with S4D, and worse at the deepest noise level”, which suggests, but
327 does not establish, that components interact rather than add. The other two
328 components tell regime stories rather than robustness stories: the convolu-
329 tional stem helps by 7.9 pp at −6 dB (removal gives 78.1 vs. 86.0; paired 95%
330 interval of the difference [−2.1, 17.9] pp, 4/5 seeds) but *costs* 21.0 pp on clean
331 cross-speed data (removal yields 98.8 vs. 77.8, with seed-std collapsing from
332 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pattern-
333 s mistransferring across conditions, or with optimisation friction under the
334 matched budget; we report both readings. Replacing the heavy-tailed state
335 parameterisation by an S4D-style one is numerically beneficial at all three
336 deep-noise levels (+2.4, +2.2, +4.2 pp), but every paired interval includes

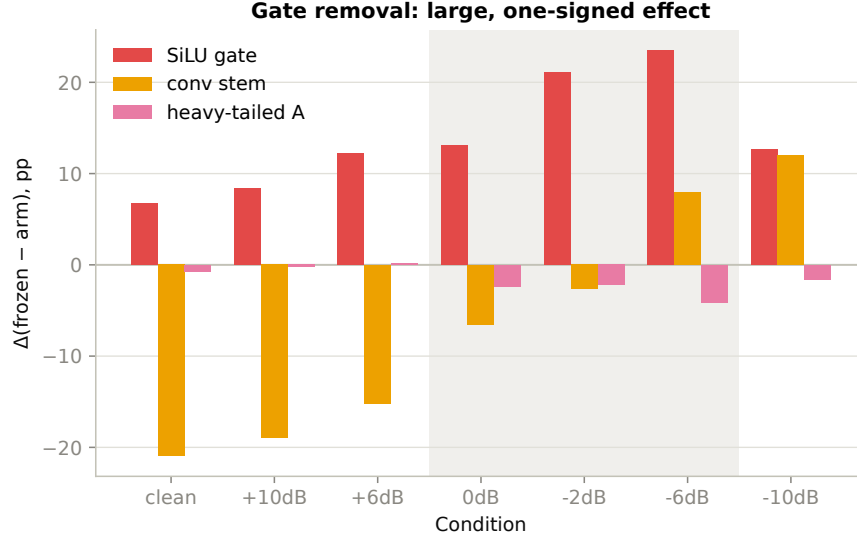


Figure 3: Component-removal Δ (percentage points, frozen – arm) by SNR for the three single-component ablations on the frozen base. Only gate removal is monotone, large, and one-signed across the adjudication band.

337 zero (e.g., $[-15.4, 7.1]$ pp at -6 dB), so we draw no directional conclusion
 338 from it (numerically, the S4D-style variant is the best deep-noise cell in the
 339 component grid, 90.1 at -6 dB).

340 4.4. Adding a narrow gate branch to S4D

Table 5: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). The S4D+gate column is the graft grid; the S4D and BM3-frozen columns, and hence R , are the component-ablation grid of Section 4.3, as pre-specified, which is why they differ from Table 2 by 0.1–0.7 pp (independent runs of the same arms).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

341 Pre-specified recovery ratio $R = \text{mean over the band of } (S4D+gate -$
 342 $S4D) / (\text{frozen} - S4D) = \mathbf{0.174}$. The pre-specified bins were $R \geq 0.7$ (con-
 343 firmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); $R =$

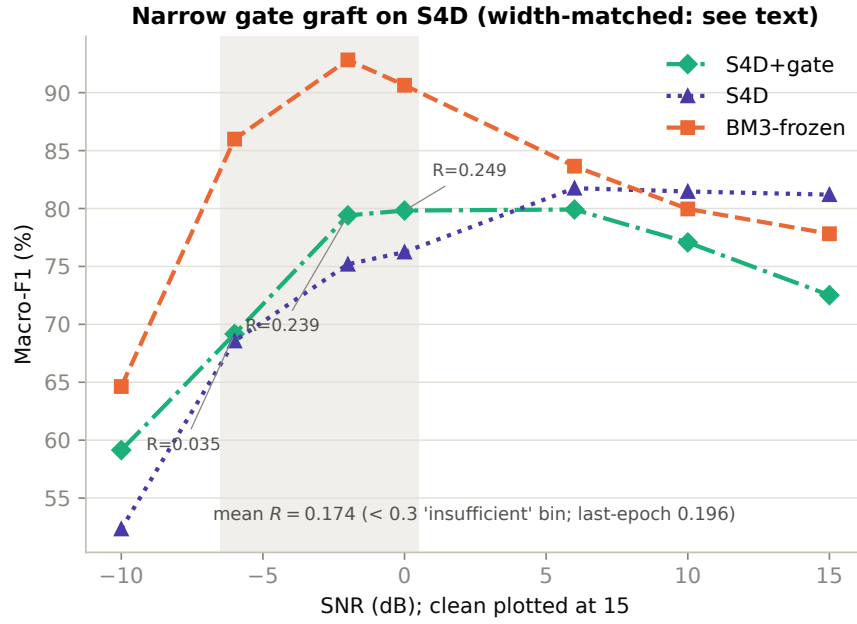


Figure 4: Graft outcome: S4D+gate, plain S4D, and BM3-frozen across the adjudication band, with the per-condition recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 6.

0.174 falls in the last. Resampling seeds within each arm gives a 95% interval of $[0.04, 0.29]$ for R , and the per-level paired gains of S4D+gate over S4D are $+3.6$ $([-2.2, 9.3])$, $+4.2$ $([1.1, 7.3])$ and $+0.6$ pp $([-2.2, 3.4])$. At the deepest adjudicated level the graft recovers essentially nothing. Removal and addition are not mirror-image interventions. In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan output at the inner width of 128 before the output projection; removing it deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch (with bias) computed from the layer input and multiplying the SSM output before the residual addition; it adds 16,640 parameters over four layers. The two share the multiplicative SiLU form but not width, projection structure or parameter count. Removal costs 13–24 pp in the frozen block whereas this graft adds 0.6–4.2 pp to S4D. What this shows is that *this gate branch, in this host and under this budget, recovers little*. Three readings fit the data and cannot be separated here: (i) host dependence, i.e. the gate needs the stem or scan context it co-trained with; (ii) implementation differences in width, projection and placement; (iii) transplant friction, i.e. the grafted branch is harder to optimise in a foreign host under the matched budget (training converges normally, but convergence is not robustness). Section 4.5 tests reading (ii) with a width-matched graft and an equal-capacity non-multiplicative control; a tuning or longer-budget comparison selected on a source-domain validation set was not run.

The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) - F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under last-epoch reporting $I = 5.2$ $[2.3, 8.1]$, 8.9 $[-6.2, 23.9]$ and 22.6 $[2.9, 42.4]$ pp at 0, -2 and -6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 $[1.4, 14.2]$, 14.4 $[-1.4, 30.3]$ and 23.1 $[8.7, 37.4]$ pp. A positive I means the gate adds more on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

4.5. A width-matched graft and an additive control

The graft of Section 4.4 is narrower and smaller than the donor gate. Two further arms rebuild it in the donor’s shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block’s inner width), the S4D layer runs on x at width 128 with state size 64—so

381 that its own parameter count and total state size equal the host’s 64-wide,
 382 state-128 layer—and an output projection maps back to width 64. *S4D+gate*
 383 (*wm*) combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same
 384 modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have
 385 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938),
 386 which biases the test towards recovery. A construction check with forward
 387 hooks confirms that the tensor entering the output projection equals the
 388 intended formula for each arm and not the other, and that the two arms are
 389 identical at initialisation. The thresholds were committed publicly before
 390 the grid ran (Section 3.5): R as before, under final-epoch reporting; the
 391 verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a
 392 bin boundary, in which case we write at the tier less favourable to the claim;
 393 and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an
 394 interval excluding zero.

Table 6: Width-matched graft and additive control on the original transition. Macro-F1
 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm
 – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and
 the narrow graft are the re-runs of Section 4.7.

Arm	0 dB	–2 dB	–6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				–0.112	[–0.26, 0.02]

395 The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159,
 396 0.298, 0.431). Its interval, [0.16, 0.52], crosses the 0.3 boundary, so the pre-
 397 specified verdict is INCONCLUSIVE between “insufficient” and “partial”,
 398 and we report it at the less favourable tier: a graft built in the donor’s shape
 399 recovers part of the gap. The recovery is not gate-specific. The additive
 400 branch, with the same parameters, recovers at least as much ($R = 0.408$),
 401 $\Delta R = -0.112$ [–0.26, 0.02], and at 0 dB the gated arm is below the additive
 402 one by 3.0 pp ([–4.5, –1.5], 0/5 seeds). With frozen—all in the denominator
 403 the gated arm’s ratio is 0.231 [0.13, 0.33]; under best-epoch reporting the two
 404 ratios are 0.459 and 0.483. What a graft recovers on this transition there-
 405 fore follows the added structure—a wider channel axis and input/output

projections—rather than the multiplicative interaction. Reading (ii) of Section 4.4 is partly confirmed, in that the narrow graft understated what added structure can recover; but in the S4D host the gate itself contributes nothing beyond an equal-capacity additive branch, whereas removing it from the frozen Mamba-3 block costs 9–23 pp under the same rule (Table 9).

4.6. The reverse transition

A pre-specified grid repeats the chain’s central contrasts on the reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen—all, BM3-frozen, frozen—gate, S4D and S4D+gate (wm), at three levels and five seeds; Section 3.1 for the data). This direction is harder for the time-invariant model: on its evaluation set, predicting the majority class gives a macro-F1 of 46.8% and uniform random guessing 41.5%, and S4D sits close to those levels (Table 7).

Table 7: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen—all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen—gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen—all − S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen − frozen—gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	−1.8
frozen—gate − S4D	[−17.6, 27.7] (2/5)	[−19.2, 23.9] (2/5)	[−18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) − S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

The first two steps replicate. The fully input-independent scan exceeds S4D by +26.3, +25.6 and +22.2 pp at 0, −2 and −6 dB, with intervals excluding zero and 5/5 seeds at every level, so by the pre-specified rule (at least two of three levels) the first step is replicated on the reverse transition. Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,

and the gateless arm is indistinguishable from S4D (intervals include zero). On the third step, which the pre-specification treats as descriptive here, the width-matched graft recovers most of the gap, $R = 0.911$ $[0.82, 1.04]$ (0.863 with frozen—all in the denominator). No additive control was run in this direction, so whether this recovery is due to the gate is not tested here; on the original transition the additive control recovered at least as much as the gate. Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

4.7. Sensitivity to the reporting rule

The tables above use best-epoch macro-F1 on the evaluation condition (Section 3.3). To measure how much this rule shapes the conclusions, we re-ran the four arms that carry the chain (BM3-frozen, S4D, frozen–gate, S4D+gate) over the adjudication band: 60 cells, 5 seeds, 50 epochs, the original training code and the same evaluation fingerprint, recording macro-F1 at every epoch and reporting it at the final epoch, where the cosine learning-rate schedule has annealed to zero. This rule uses no evaluation data to pick a model. The re-run used a different PyTorch/Triton build from the original grids; its best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp (the largest deviation is in the high-variance frozen–gate arm at -2 dB). The S4D-wide capacity control and the conv-stem and A -parameterisation ablations were not re-run; the original BM3 arm was re-run in the stage-2 grid of Section 4.2, so $\Delta(\text{BM3} - \text{frozen})$ is available under both rules.

Table 8: Macro-F1 (%), mean over 5 seeds, under best-epoch reporting (re-run) and final-epoch reporting.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
frozen–gate	77.7	74.0	73.1	71.7	61.9	60.5
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Final-epoch values are lower than best-epoch values by 1.4–9.5 pp across the twelve arm-by-level means, so the absolute levels in Tables 2–5 are opti-

mistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0, -2 , -6 dB), and the contrasts that carry the argument are correspondingly stable (Table 9).

Table 9: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	-2 dB	-6 dB
	+15.5	+19.8	+16.4
BM3-frozen $-$ S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen $-$ frozen-gate	[2.9, 15.3] (5/5)	$[-0.8, 32.0]$ (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	-6.2
frozen-gate $-$ S4D	$[-2.9, 15.8]$ (4/5)	$[-14.5, 22.9]$ (3/5)	$[-23.5, 11.2]$ (1/5)
	+3.9	+6.8	-0.1
S4D+gate $-$ S4D	$[-0.8, 8.6]$ (4/5)	[2.4, 11.1] (5/5)	$[-4.0, 3.9]$ (2/5)

Selectivity. The frozen block’s advantage over S4D is +15.5, +19.8 and +16.4 pp (and +21.9, +24.0, +21.2 pp for the fully frozen scan, Section 4.2), versus +14.6, +17.2 and +17.1 pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified threshold at every level; the interval at -2 dB includes zero ($[-0.8, 32.0]$ pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at -2 dB). *Gateless versus S4D.* Under final-epoch reporting the gateless arm is not below S4D at 0 and -2 dB (point estimates +6.4 and +4.2 pp) and is below it at -6 dB (-6.2 pp); all three intervals include zero. The statement that gate removal “falls below S4D” is therefore supported only as a point-estimate observation at the deepest level, and we phrase it as “to or below” elsewhere. *Graft.* The recovery ratio is 0.249, 0.341 and -0.003 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point estimate. The seed-resampled 95% interval is $[0.09, 0.31]$ and so reaches the 0.3 boundary: with five seeds the data support “far below confirmation (0.7) and probably below 0.3”, not a precise value of R . The graft gain over S4D is resolved at -2 dB only (+6.8 pp, $[2.4, 11.1]$) and is indistinguishable from zero at -6 dB.

Because the final epoch of the same evaluation condition has been in-

476 spected repeatedly during this study, last-epoch values are also not a fresh
 477 confirmatory test. The reporting rule therefore changes absolute levels and
 478 the strength of the gateless-versus-S4D statement, but not the ordering BM3-
 479 frozen > {S4D+gate, frozen–gate, S4D}, the size of the frozen block’s ad-
 480 vantage over S4D, or the verdicts of the pre-specified gate and graft rules.

481 4.8. The interaction effect concentrates on the hard class

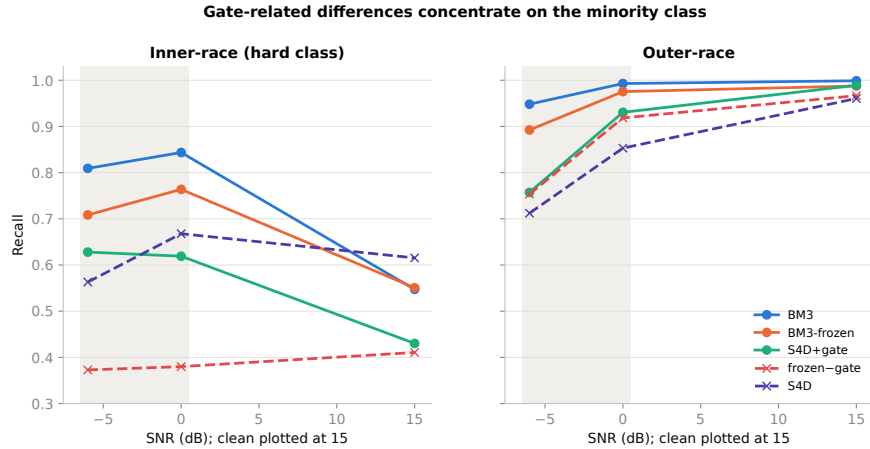


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

482 Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch
 483 selection on the evaluation condition: 5 arms $\times$ {clean, 0, -6 dB} $\times$ 5 seeds,
 484 per-sample predictions dumped and re-verified against reported metrics cell-
 485 by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62
 486 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate
 487 presence: in all three gated arms IR recall at -6 dB exceeds its clean val-
 488 ue (e.g., 0.55 $\rightarrow$ 0.81 for BM3), while in both gateless arms it is below its
 489 clean value (frozen–gate: 0.41 $\rightarrow$ 0.37; S4D: 0.62 $\rightarrow$ 0.56, non-monotone,
 490 rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—
 491 gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from
 492 predictions pooled over seeds), and the graft’s noise-time gains also concen-
 493 trate on IR (-6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages
 494 the two classes, the arm-level robustness gaps of Sections 4.1–4.4 are carried
 495 substantially by the hard class. The gate-related differences are therefore

class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a “low-margin” class. Because this grid is separate from the main 50-epoch grid and the IR class comes from a single training bearing (Section 3.1), these results are descriptive and do not decompose the main-grid effects.

5. Discussion

What can be credited? The ablation chain supports a narrow, conditional statement: within this Mamba-3 block, on this testbed and with noise-aware training, (i) making the scan fully input-independent—first the Δ , A , B , C projections, then the trapezoidal weight and rotation angles as well—did not remove the block’s deep-noise advantage over a time-invariant S4D, and at the deepest level the fully frozen arm matched or exceeded the original block with fewer parameters, (ii) removing the output gate removed most of that advantage—(i) and (ii) holding on the reverse transition as well—and (iii) adding the gate to S4D recovered part of the advantage only together with added width and projections, and an additive branch of the same size recovered as much. These observations are consistent with a host-dependent gate benefit. They do not identify which other component the gate depends on. Nor do they separate “input-dependence contributes nothing here” from “what it contributes is recovered by learned constants under this budget”; the two readings are indistinguishable in our data and lead to the same attribution. Representability results for selectivity [8] and the S4D-superiority finding on clean data [4] are not contradicted by a single ablation on one testbed; our clean-data parity agrees with the latter, and our deep-noise result suggests that in this block the output gate, rather than any input-dependent term of the scan, accounts for most of the measured advantage.

Implications for backbone selection and for transplant practice.

For practitioners choosing backbones for noisy, drifting signal regimes, the results suggest that the block’s benefit should be judged as a composition, and that a gate added to a different host should not be expected to reproduce it: in our S4D host a gate built in the donor’s shape recovered no more than an additive branch of the same size (Section 4.5), and neither design was tuned. A hypothesis for future work follows: a recomposed block guided by the component map (no conv stem, gate retained, S4D-style A) may match or exceed the original on this testbed; we leave it untested, because post-hoc

recomposition after seeing the map would not carry the evidential status of the pre-specified chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate-related differences concentrate on the minority, lowest-recall class (Section 4.8), and matched-noise training lifts that class *only when the gate is present*—suggesting the gate is the component through which matched-noise training is converted into hard-class margin, i.e., an exploiter of noise-aware training rather than a test-time denoiser (this reading is bounded by the matched-SNR scope, Section 6). Second, its role appears closer to adaptive per-sample feature suppression than to generic denoising. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the weakest of the three observations, since the gateless–S4D gap is not resolved with five seeds (Section 4.7). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed scope.** All claims are bounded to bearing vibration from one pair of operating conditions of one dataset (37.5 Hz/11 kN and 40 Hz/10 kN; speed and load both change), binary classification, GPU execution (no edge/real-time claims). The pair is run in both directions, but the reverse direction uses the same eight bearings with their roles swapped and has a single IR bearing in its evaluation set, so it is not an independent replication; the third condition of the dataset has no IR bearing. The five seeds do not resample bearings (Section 3.1). Independent bearings, multi-class settings and a second dataset are future work.
2. **Attribution level.** Credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed, and the host-by-gate contrast of Section 4.4 is a four-arm estimate that cannot be assigned to a specific component.

- 566 3. **The transplant conclusion is budget-conditional.** All arms share
567 a matched 50-epoch budget; the grafted gate trains normally yet re-
568 covers little, and convergence is not robustness (Section 4.4)—but we
569 cannot exclude that a longer or graft-tuned budget would recover more.
570 Our statement is therefore limited to *this gate branch, in this host, un-*
571 *der a matched training budget.*
- 572 4. **Matched-SNR scope.** Training and evaluation share the same in-
573 jected SNR throughout (Section 3.1); “robustness” in this paper is
574 matched-noise robustness under noise-aware training, not clean-train/noisy-
575 test generalisation. The gate’s benefit may operate by exploiting matched-
576 noise training (consistent with Section 4.8) rather than by test-time de-
577 noising; attribution under mismatched-noise protocols is untested and
578 may differ.
- 579 5. **Graft dual reading.** Interaction dependence vs. transplant friction
580 (Section 4.4), the latter now subsumed under the budget condition of
581 item 3.
- 582 6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a
583 TCN baseline slightly overtakes the Mamba-style block, and -10 dB
584 is near-chance for the time-invariant baseline and excluded from ad-
585 judication; the demonstrated advantage regime is matched, Gaussian-
586 dominated noise.
- 587 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
588 mistransfer and optimisation-difficulty readings, reported in place (Sec-
589 tion 4.3).
- 590 8. **Reporting rule.** Reported values are best-epoch macro-F1 on the
591 evaluation condition, with no separate validation split (Section 3.3);
592 absolute levels are optimistic by 1.4–9.5 pp in the arms re-run under
593 final-epoch reporting (Section 4.7). That re-assessment covers the arms
594 that carry each step at the three adjudication levels (the BM3 arm was
595 re-run in the grid of Section 4.2); the capacity control, the stem and A
596 ablations, the $+10/+6/\text{clean}/-10$ dB levels and the per-class grid were
597 not re-run, and no source-domain validation rule was available. The
598 final epoch is a fixed-budget rule, not a tuned one.
- 599 9. **Scope of the freezing intervention.** The chain’s first step freezes
600 the input-dependent parts of Δ , A , B , C only; the fully input-independent
601 scan is a separate grid (Section 4.2) on the three adjudication levels,
602 which was specified after the earlier grids had been seen. No equivalence
603 margin was pre-specified for the cost of freezing, so “no measurable

cost” means the point estimates favour the frozen arms with intervals including zero, not a demonstrated equivalence. Freezing is implemented inside this Mamba-3 layer; a conclusion about input-dependence in general would need other blocks and tasks.

10. **Asymmetric gate interventions.** The width-matched graft and its additive control (Section 4.5) were run on the original transition only; on the reverse transition the graft has no additive control. In the Mamba-3 block the gate was removed, not replaced by an additive branch of the same size, so the two hosts are still not compared under identical interventions, and the width-matched arms are larger than the donor block.

11. **Strength of the pre-specification evidence.** Thresholds were pre-specified internally; for all but the last follow-up grid the supporting evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.

12. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.

13. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled drift $\times$ noise testbed with matched-noise training, freezing the input-dependent Δ , A , B , C projections costs at most about 5 pp (an estimate that five seeds leave imprecise) while 14–20 pp of advantage over a time-invariant SSM remains, and freezing the trapezoidal weight and rotation angles as well—leaving no input-dependent

638 tensor in the scan—costs nothing measurable and leaves 21–24 pp of advan-
639 tage with fewer parameters. The multiplicative output gate is the one ablated
640 component whose removal causes a large, one-signed loss (9–24 pp, returning
641 the block to roughly the level of plain S4D or below), yet adding the gate to
642 S4D recovers no more than the added structure does: 17–20% of the gap for
643 a narrow graft, about 30% for one built in the donor’s width, and at least
644 as much for an additive branch of the same size. We read this as a host
645 dependence of the gate’s benefit in this design, concentrated on the minority
646 class. The first two steps replicate on the reverse transition. The evidence
647 rests on one pair of operating conditions run in both directions with the same
648 eight bearings, five seeds, and an evaluation-condition epoch-selection rule
649 that we re-assessed for the arms that carry each step. We offer the three-step
650 chain—freeze, remove, add—as a protocol for turning architecture compar-
651 isons into component-level tests, with the caveat that the interventions must
652 be matched in width and capacity for the add step to be conclusive, and
653 we release our adjudication trail, including the incidents in which automated
654 adjudication departed from the pre-specified criteria.

655 **CRedit authorship contribution statement**

656 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
657 analysis, Investigation, Data curation, Writing – original draft, Writing –
658 review & editing, Visualization, Supervision, Project administration.

659 **Declaration of competing interest**

660 The authors declare that they have no known competing financial inter-
661 ests or personal relationships that could have appeared to influence the work
662 reported in this paper.

663 **Declaration of Generative AI and AI-assisted technologies in the** 664 **writing process**

665 During the preparation of this work the authors used LLM-based agents
666 to execute and cross-check pre-specified analysis and adjudication passes as
667 described in Appendix C, and to assist with manuscript drafting and figure
668 generation. After using these tools, the authors reviewed and edited the con-
669 tent as needed and take full responsibility for the content of the published

670 article. All numeric thresholds, adjudication rulings, and scientific interpre-
671 tations reported in this paper were fixed and made by the human authors,
672 as detailed in Appendix C.

673 Data availability

674 The code, pre-specification artifacts, and adjudication trail for this study
675 are available at <https://github.com/jeff-1980/bm3-jizi>. The repository
676 includes the two follow-up grids of Sections 4.2 and 4.7 (120 cells with
677 per-epoch curves, their drivers, the layered-freeze arms and their construc-
678 tion check) and the harness’s own imported modules, so the training entry
679 point runs from a fresh checkout.

680 Appendix A. Fairness protocol

681 *Appendix A.1. Evaluation-set fingerprint*

682 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
683 256 hash of the raw held-out test-label byte array, computed once per run
684 and asserted identical across every arm/condition/seed cell within a grid.
685 It fingerprints *that the same held-out label vector was scored every time*, not
686 model weights or predictions. It is also the split on which the best-epoch rule
687 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
688 five grids is:

689 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

690 This value is recomputed independently at adjudication time (not merely
691 copied from the run log) and cross-checked row-by-row against every grid’s
692 fairness report before any comparison across arms is treated as valid.

693 *Appendix A.2. Per-arm parameter counts*

694 Table A.10 lists every arm’s parameter count, each obtained directly via
695 `sum(p.numel() for p in model.parameters())` on the instantiated model
696 (not estimated), cross-verified against an independently maintained archi-
697 tecture map.

Table A.10: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

Table A.11: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (mtime 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

698 *Appendix A.3. Five-batch pre-specification timeline*

699 Table A.11 lists, for each of the five pre-specified grids, the artifact that
700 fixed its adjudication thresholds, that artifact’s creation/modification date,
701 and the corresponding grid’s run-directory start time. In every case the
702 prereg artifact predates the grid it governs.

703 Two honest caveats. First, the baseline batch predates the orchestration
704 infrastructure used for the other four batches by five days; its prereg trail
705 rests on the task card’s creation timestamp alone, not on the same automat-
706 ed executor/reviewer machinery documented in Appendix C, and we state
707 this rather than implying a uniform process across all five batches. Second,
708 for the component-ablation grid, two additional JSON files bearing “pre-
709 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
710 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
711 time; both are explicitly marked void in the corresponding decision artifact’s
712 `void_preregs` field, and neither was used to adjudicate any result in this pa-
713 per. They are the on-disk residue of the first post-hoc criterion substitution
714 described in Appendix C, retained here (rather than deleted) as part of the
715 released adjudication trail.

716 **Appendix B. Confusion matrices and a repeat-run check**

717 *Appendix B.1. Full confusion-matrix set*

718 Table B.12 gives the complete confusion-matrix set underlying the per-
719 class analysis of Section 4.8: 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
720 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
721 counts were dumped per-sample and re-verified against the reported recal-
722 l/precision/F1 metrics cell-by-cell before use.

Table B.12: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
S4D+gate	−6 dB	15,634	1,886	2,543	6,177
	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
frozen−gate	−6 dB	13,267	4,253	3,244	5,476
	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
S4D	−6 dB	13,200	4,320	5,469	3,251
	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

723 Appendix B.2. Repeat-run check

724 Because two independent process runs of an identical configuration are
725 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
726 ductions in the selective-scan and normalisation ops), we measured a process-
727 level noise floor directly: the same arm, condition, and seed (`bm3.frozen`,
728 clean, seed 0, 2 epochs) was run twice in separate processes with identical
729 evaluation fingerprints and identical parameter counts confirmed. The two
730 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

731 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
732 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
733 run, and therefore neither an estimate nor a bound of process variability for
734 full-length runs. It is an order of magnitude below the seed-level variabil-
735 ity observed within the deep-noise band across the curve tables underlying
736 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
737 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
738 are judged against seed-level variability, not against this single repeat.

739 Appendix C. Adjudication incidents, automation disclosure, and 740 human adjudication gates

741 This appendix gives the material Section 3.5 summarises in one sentence:
742 what the pre-specification discipline actually caught, how the analysis/review

743 pipeline was implemented, and exactly where human judgement entered. We
 744 report this in full rather than in summary because the growing use of au-
 745 tomation in pre-specified analysis pipelines raises exactly the failure modes
 746 documented below, and because the reader should be able to audit, not mere-
 747 ly take on faith, that no post-hoc criterion change survived into the results
 748 reported in Section 4.

749 *Appendix C.1. Automation disclosure*

750 The mechanical stages of this study’s analysis pipeline — applying pre-
 751 specified numeric thresholds to aggregated results and emitting a decision
 752 artifact (the “executor” role), and independently re-computing aggregates,
 753 re-verifying evaluation fingerprints row-by-row, and checking the executor’s
 754 decision against the pre-specified rule (the “reviewer” role) — were imple-
 755 mented using LLM-based coding agents operating under an orchestration
 756 harness, not by hand-written deterministic scripts alone. We state this plain-
 757 ly: the executor and reviewer passes described throughout Section 3.5 and in
 758 the three incidents below (Appendix C.2) are automated agent passes, and
 759 the incidents are, in each case, failures or near-failures of that automation.
 760 We consider this disclosure a necessary complement to the pre-specification
 761 protocol itself, not a caveat that weakens it: the protocol was designed on
 762 the assumption that automation of this kind is fallible, and its value lies in
 763 the adjudication gates of Appendix C.3, which the incidents below show
 764 were load-bearing rather than decorative.

765 *Appendix C.2. Three incidents*

766 *Incident 1 — post-hoc criterion substitution, frozen-selectivity grid (2026-*
 767 *07-03)..* The pre-specified rule for the necessity test (Section 4.1) was: at-
 768 tribute robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in the deep-
 769 noise band, with the frozen arm collapsing toward S4D. The executor’s
 770 automated decision instead wrote verdict `SELECTIVITY_CLAIM_FALSIFIED`
 771 using an unpre-specified criterion — “BM3 does not significantly outper-
 772 form BM3-frozen in a majority of conditions” — substituted for the nu-
 773 meric threshold. On human review of the decision artifact, this substitu-
 774 tion was caught before any downstream write used it: the four measured
 775 deltas were 5.14/3.34/2.19/3.80 pp, all below the pre-specified 10 pp floor,
 776 which correctly resolves to pre-specified outcome (b), “mechanism not prima-
 777 ry” (`SELECTIVITY_NOT_PRIMARY_MECHANISM`), a materially different and more
 778 specific claim than “falsified.” A correction card mechanically re-applied the

779 original pre-specified rule and overwrote the erroneous decision artifact, with
780 the human ruling and its justification timestamped in the artifact itself.

781 *Incident 2 — post-hoc criterion substitution, component-ablation grid (2026-*
782 *07-04)..* During adjudication of the single-component-removal grid (Section 4.3),
783 the executor abandoned the pre-specified percentage-point threshold and
784 self-authored a “mean $\pm$ std interval non-overlap” heuristic, embodying it
785 in JSON files (`secondary_prereg_20260704-0708` and `secondary_deltapp-`
786 `prereg_20260704-0719`) generated *after* the grid had already run — i.e., ar-
787 tifacts presented as if they were the pre-specification but in fact post-dating
788 the data they purported to govern. Separately, and independent of this sub-
789 stitution, the true pre-specification existed on a task-tracking card created
790 2026-07-03 (before the grid started at 10:34) but had not been copied into
791 the working directory, so an independent reviewer pass, unable to see the
792 card, correctly flagged “no pre-specification precedes this data” and pro-
793 posed a full grid rerun as remedy. Human adjudication rejected the rerun:
794 the provenance chain was intact (the true prereg predated the grid), the de-
795 fect was a materialization gap, not a genuine pre-specification violation. The
796 threshold-substitution defect was corrected by mechanically re-applying the
797 true, now-materialized threshold (documented in `prereg_provenance.md`);
798 the two post-hoc JSON files were not deleted but are explicitly marked void
799 (`void_preregs`) in the corresponding decision artifact and are retained as
800 part of the released trail (Appendix A.3). The resulting adjudication — gate
801 removal as the only ablated component clearing the primary-driver threshold
802 — is the one reported in Section 4.3.

803 *Incident 3 — stale-artifact false verification, graft grid (2026-07-05)..* A
804 stage-1 task card for the graft construction (Section 4.4) was marked PASS
805 despite no implementation work having occurred. Two compounding caus-
806 es were identified on audit: (a) the card’s task section was headed `## TASK`
807 (`stage-1 scope`) rather than the bare `## TASK` string the automation matched
808 exactly, so the executor received an empty task body and did no work, log-
809 ging this fact accurately; (b) the reviewer’s verification rubric described re-
810 quirements generically (“scripts present”) rather than naming specific files,
811 so the reviewer verified against unrelated leftover artifacts from the prior
812 component-ablation batch (a different grid’s unit-check and smoke-test out-
813 puts) and passed the card without those artifacts having any bearing on the
814 graft arm. At the time of the false PASS, the grafted arm (`s4d.plus.gate`)

815 had no implementation anywhere in the repository — confirmed by a zero-hit
816 search across the codebase. This was caught by an independent audit, not
817 by the pipeline itself; the false PASS was manually voided, and the card was
818 re-issued with three structural fixes: bare section headers only, a rubric that
819 names exact files rather than generic requirements, and an “anti-staleness
820 clause” requiring every verification claim to cite a file path and a modifica-
821 tion time newer than a stated cutoff. The graft grid was then executed from
822 scratch under the corrected card; the numbers reported in Section 4.4 and Ta-
823 ble 5 come from that later, correctly verified run, not from the falsely-passed
824 stage.

825 *Appendix C.3. Human adjudication gates*

826 Independent of which stages were automated, the following were, without
827 exception, fixed or decided by the human authors, not by any agent:

828 **All numeric adjudication thresholds** (the 10 pp necessity floor in
829 Section 4.1, the ≥ 8 pp primary-driver floor and $|\Delta| < 3$ pp not-a-driver
830 floor in Section 4.3, the $R \geq 0.7/0.3 \leq R < 0.7/R < 0.3$ graft recovery
831 bands in Section 4.4) were written to disk before the corresponding grid
832 was run, as documented in Appendix A.3.

833 **All three rulings in Appendix C.2** — rejecting Incident 1’s sub-
834 stituted criterion, rejecting Incident 2’s rerun remedy while accepting
835 that a real (if unmaterialized) pre-specification existed, and voiding
836 Incident 3’s false PASS — were made by the human authors on direc-
837 t inspection of the underlying artifacts (decision JSON, task-tracking
838 card timestamps, repository search results), not by the executor or
839 reviewer agents that produced or reviewed those artifacts.

840 **All interpretive claims in Section 5** (the composition-bound read-
841 ing of gating, the dual readings offered for the conv-stem cost and the
842 graft failure) are authors’ interpretations of adjudicated numeric out-
843 comes, not agent-generated conclusions.

844 **The decision to report, rather than suppress, all three inci-**
845 **dents**, including the two instances where automation initially produced
846 an incorrect scientific verdict, was made by the human authors as a
847 condition of using automation in this pipeline at all.

848 No agent in this pipeline was permitted to overwrite a protected decision
849 artifact, alter a pre-specified threshold, or mark its own output as final; every
850 one of the three incidents above was caught before the erroneous artifact
851 propagated into a reported number in Section 4.

852 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

853 Elsevier’s policy on generative AI and AI-assisted technologies requires
854 that any such use in the writing process be disclosed, that AI tools not be
855 listed as authors, and that human authors retain full responsibility for the
856 content and its accuracy. The declaration accompanying this manuscript
857 (main text, “Declaration of Generative AI and AI-assisted technologies in
858 the writing process”) satisfies this requirement for the manuscript text itself.
859 We additionally extend disclosure, on our own initiative, to the experimental
860 adjudication pipeline described in this appendix, on the view that trans-
861 parency about automation matters most exactly where it could affect the
862 reported scientific conclusions, not only the prose describing them. The hu-
863 man adjudication gates of [Appendix C.3](#) are the mechanism by which we
864 take the responsibility the policy assigns to authors: every threshold, every
865 ruling on a caught defect, and every interpretive claim in this paper was
866 fixed or decided by a human author, with the automation’s role confined to
867 mechanical computation subject to audit and override.

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